Notes on Profile Composite Likelihood

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1 Background: Composite Likelihood

1.1 Motivation

The full joint likelihood $L(\boldsymbol{\theta}; \mathbf{y})$ may be unavailable, intractable, or computationally prohibitive in models with:

- large dense covariance matrices (e.g., spatial/geostatistical models),
- high-dimensional integrals over random effects,
- dependent data where specifying the full joint density is difficult.

A **composite likelihood** sidesteps these issues by forming a surrogate objective from lower-dimensional marginal or conditional densities, which are individually tractable even when the joint is not.

1.2 Definition

Let $\mathbf{y}$ be observed data and $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^d$ the parameter vector. Given a collection of events or sub-vectors indexed $k = 1, \ldots, K$, each with associated likelihood contribution $L_k(\boldsymbol{\theta}; \mathbf{y})$ derived from a marginal or conditional density, the **composite likelihood function** is

$$CL(\boldsymbol{\theta}; \mathbf{y}) = \prod_{k=1}^{K} L_k(\boldsymbol{\theta}; \mathbf{y})^{w_k},$$

with nonnegative weights w_k. Working on the log scale, the **composite log-likelihood** is

$$\ell_C(\boldsymbol{\theta}) = \sum_{k=1}^{K} w_k \log L_k(\boldsymbol{\theta}; \mathbf{y}).$$

Common special cases include:

- **Pairwise (marginal) likelihood**: L_k is the joint density of a pair (Y_i, Y_j); introduced formally in Cox & Reid (2004).
- **Conditional composite likelihood**: L_k is a conditional density $f(y_i \mid y_j)$.
- **Independence likelihood**: $L_k = f(y_i; \boldsymbol{\theta})$, ignoring dependence entirely.

Lindsay (1988) introduced the general framework and established foundational consistency and asymptotic normality results.

1.3 The Godambe Information and Sandwich Variance

A critical difference from ordinary likelihood is that the **second Bartlett identity fails**: in general,

$$H(\boldsymbol{\theta}) \neq J(\boldsymbol{\theta}),$$

where

$$H(\boldsymbol{\theta}) = -\mathbb{E}\big[\nabla^2_{\boldsymbol{\theta}} \ell_C(\boldsymbol{\theta})\big] \quad \text{(sensitivity matrix)},$$
$$J(\boldsymbol{\theta}) = \mathrm{Var}[\nabla_{\boldsymbol{\theta}} \ell_C(\boldsymbol{\theta})] \quad \text{(variability matrix)}.$$

The **Godambe information matrix** (analogous to Fisher information) is

$$G(\boldsymbol{\theta}) = H(\boldsymbol{\theta})\, J(\boldsymbol{\theta})^{-1}\, H(\boldsymbol{\theta}).$$

The maximum composite likelihood estimator $\hat{\boldsymbol{\theta}}_{CL}$, obtained by solving $\nabla_{\boldsymbol{\theta}} \ell_C(\boldsymbol{\theta}) = \mathbf{0}$, satisfies

$$\sqrt{n}\,(\hat{\boldsymbol{\theta}}_{CL} - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}\big(\mathbf{0},\ G(\boldsymbol{\theta}_0)^{-1}\big)\,,$$

so the asymptotic variance is the *inverse* Godambe information, not simply H^{-1} as it would be under a correctly specified full likelihood. This is the **sandwich variance** structure familiar from Godambe's estimating equations framework and from quasi-likelihood theory.

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2 Profile Composite Likelihood

2.1 Setup and Definition

Partition the parameter as $\boldsymbol{\theta} = (\boldsymbol{\psi}, \boldsymbol{\lambda})$, where $\boldsymbol{\psi} \in \mathbb{R}^p$ is the **parameter of interest** and $\boldsymbol{\lambda} \in \mathbb{R}^q$ is a **nuisance parameter**.

The **profile composite log-likelihood** for $\boldsymbol{\psi}$ is

$$\ell_{C,p}(\boldsymbol{\psi}) = \ell_C\Big(\boldsymbol{\psi},\ \hat{\boldsymbol{\lambda}}_{\boldsymbol{\psi}}\Big)\,,$$

where

$$\hat{\boldsymbol{\lambda}}_{\boldsymbol{\psi}} = \arg\max_{\boldsymbol{\lambda}}\ \ell_C(\boldsymbol{\psi}, \boldsymbol{\lambda})$$

is the constrained maximizer of the composite log-likelihood with $\boldsymbol{\psi}$ fixed.

This is structurally identical to the ordinary profile likelihood $\ell_p(\boldsymbol{\psi})$: we concentrate out the nuisance parameters by replacing them with their conditional composite MLEs.

2.2 Why Ordinary Profile Likelihood Theory Fails

In the full-likelihood case, the profile log-likelihood ratio

$$W(\boldsymbol{\psi}_0) = 2\left[\ell(\hat{\boldsymbol{\theta}}) - \ell_p(\boldsymbol{\psi}_0)\right] \xrightarrow{d} \chi^2_p$$

under $H_0 : \boldsymbol{\psi} = \boldsymbol{\psi}_0$. This Wilks-type result depends on the second Bartlett identity holding. Because composite likelihoods are not genuine likelihoods, $H \neq J$ in general, and the analogous ratio

$$W_C(\boldsymbol{\psi}_0) = 2\left[\ell_C(\hat{\boldsymbol{\theta}}_{CL}) - \ell_{C,p}(\boldsymbol{\psi}_0)\right]$$

does **not** converge to χ^2_p. Naive use of W_C for inference—confidence regions, likelihood ratio tests—leads to incorrect coverage and type I error.

2.3 Asymptotic Distribution of the Profile Composite Likelihood Ratio

Under regularity conditions, the unadjusted statistic $W_C(\boldsymbol{\psi}_0)$ converges to a weighted sum of independent χ^2_1 random variables:

$$W_C(\boldsymbol{\psi}_0) \xrightarrow{d} \sum_{j=1}^{p} \kappa_j Z_j^2, \quad Z_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1),$$

where $\kappa_1, \ldots, \kappa_p$ are eigenvalues of a matrix that depends on the profile Godambe information. This motivates a rescaling to recover a standard χ^2_p limit.

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3 Adjusted Composite Likelihood Ratio Statistics

3.1 The Rescaled Statistic

Write the full parameter Godambe information in block form corresponding to $(\boldsymbol{\psi}, \boldsymbol{\lambda})$:

$$G(\boldsymbol{\theta}) = H(\boldsymbol{\theta})\, J(\boldsymbol{\theta})^{-1}\, H(\boldsymbol{\theta}).$$

Partition H and J conformably:

$$H = \begin{pmatrix} H_{\psi\psi} & H_{\psi\lambda} \\ H_{\lambda\psi} & H_{\lambda\lambda} \end{pmatrix}, \quad J = \begin{pmatrix} J_{\psi\psi} & J_{\psi\lambda} \\ J_{\lambda\psi} & J_{\lambda\lambda} \end{pmatrix}.$$

The **profile Godambe information** for $\boldsymbol{\psi}$ is the $(\boldsymbol{\psi}, \boldsymbol{\psi})$ block of G^{-1} inverted—i.e., the Schur complement analog using Godambe rather than Fisher information. Let $G_{\psi\psi\cdot\lambda}$ denote the profile Godambe information (the appropriate Schur complement of G in $\boldsymbol{\lambda}$).

The adjusted composite likelihood ratio statistic proposed by Geys, Molenberghs & Ryan (1999) is

$$\boxed{W_C^*(\boldsymbol{\psi}_0) = \frac{p}{\operatorname{tr}\left(\hat{C}\right)}\, W_C(\boldsymbol{\psi}_0)},$$

where $\hat{C}$ is the estimated matrix whose eigenvalues govern the asymptotic distribution. Under H_0, $W_C^* \xrightarrow{d} \chi^2_p$.

3.2 Wald-Type Inference

A simpler route that avoids adjusting the ratio statistic is to use the **Wald statistic** directly from the sandwich variance:

$$T_W(\boldsymbol{\psi}_0) = n\,(\hat{\boldsymbol{\psi}}_{CL} - \boldsymbol{\psi}_0)^\top \hat{G}_{\psi\psi\cdot\lambda}\,(\hat{\boldsymbol{\psi}}_{CL} - \boldsymbol{\psi}_0) \xrightarrow{d} \chi^2_p,$$

where $\hat{G}_{\psi\psi\cdot\lambda}$ is the estimated profile Godambe information evaluated at $\hat{\boldsymbol{\theta}}_{CL}$. This is the most commonly implemented approach in practice, since it requires only the sandwich variance from the full parameter estimate, without profiling at all.

3.3 Score-Type (Rao) Statistic

An analog to the score test is

$$T_S(\boldsymbol{\psi}_0) = \nabla_{\boldsymbol{\psi}} \ell_{C,p}(\boldsymbol{\psi}_0)^\top \hat{G}^{-1}_{\psi\psi\cdot\lambda}\, \nabla_{\boldsymbol{\psi}} \ell_{C,p}(\boldsymbol{\psi}_0) \xrightarrow{d} \chi^2_p,$$

evaluated only under the null, which can be useful when the unconstrained maximization of the composite likelihood is expensive.

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4 Nuisance Parameters and Information Bias

4.1 The Information Bias Problem

Following Lindsay (1982), a composite likelihood is called **information-unbiased** if $H(\boldsymbol{\theta}) = J(\boldsymbol{\theta})$, and **information-biased** otherwise. Information bias has a direct analogue to the well-known issue with ordinary profile likelihood:

- In ordinary profile likelihood, profiling introduces an $O(1)$ bias in the profile score that does not vanish, motivating modified profile likelihoods (Barndorff-Nielsen 1983; Cox & Reid 1987).
- In profile composite likelihood, the analogous bias is compounded: we have both the information-bias inherent to ℓ_C *and* the additional bias from plugging in $\hat{\boldsymbol{\lambda}}_{\boldsymbol{\psi}}$.

4.2 Consequences for Efficiency

A perhaps surprising result (Xu & Reid, 2011; Mardia et al., 2009) is that information-biased composite likelihoods can be *more efficient* in the presence of nuisance parameters, because the profile Godambe information $G_{\psi\psi\cdot\lambda}$ can exceed what a naive Schur complement of H would give. Efficiency comparisons require working with G directly rather than H alone.

4.3 Many Nuisance Parameters: The Neyman–Scott Problem

When the number of nuisance parameters grows with n (the incidental parameters problem), profiling in ordinary likelihood is known to be inconsistent (Neyman & Scott, 1948). Composite likelihood can help here. Gao & Song (2013) consider estimation of a fixed-dimensional $\boldsymbol{\psi}$ with diverging-dimensional $\boldsymbol{\lambda}$: by profiling lower-dimensional marginal composite likelihoods, one can recover $\sqrt{n}$-consistent estimation of $\boldsymbol{\psi}$ even as $q \to \infty$ with n.

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5 Confidence Regions

5.1 From the Adjusted Ratio

A $(1-\alpha)$ confidence region for $\boldsymbol{\psi}$ based on the adjusted ratio is

$$\mathcal{C}_{1-\alpha} = \left\{\boldsymbol{\psi} : W_C^*(\boldsymbol{\psi}) \le \chi^2_{p,\,1-\alpha}\right\}.$$

For a scalar parameter of interest ($p = 1$), this gives a confidence interval by finding the roots of $W_C^*(\psi) = \chi^2_{1,\,1-\alpha}$.

5.2 From the Wald Statistic

$$\mathcal{C}^W_{1-\alpha} = \left\{\boldsymbol{\psi} : T_W(\boldsymbol{\psi}) \le \chi^2_{p,\,1-\alpha}\right\} = \hat{\boldsymbol{\psi}}_{CL} \pm z_{1-\alpha/2}\sqrt{\hat{G}^{-1}_{\psi\psi\cdot\lambda}/n} \quad (p = 1).$$

In most software implementations (e.g., `lavaan`, spatial composite likelihood packages), the standard errors reported are $\sqrt{\mathrm{diag}(\hat{G}^{-1}/n)}$—that is, the square root of the diagonal of the inverse Godambe information—and confidence intervals are formed as Wald intervals.

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6 Summary of Key References

Reference	Contribution
Lindsay (1988)	Founding framework; consistency and asymptotic normality of MCLE
Cox & Reid (2004)	Pairwise marginal likelihood; formal composite likelihood setup
Geys, Molenberghs & Ryan (1999)	Adjusted composite likelihood ratio statistic
Varin (2008)	Review of composite marginal likelihoods
Varin, Reid & Firth (2011)	Comprehensive overview; Godambe information, efficiency, nuisance parameters
Xu & Reid (2011)	Information bias, efficiency in the presence of nuisance parameters
Gao & Song (2013)	Profiling with many nuisance parameters; incidental parameters problem
Barndorff-Nielsen (1983); Cox & Reid (1987)	Modified/adjusted profile likelihood (ordinary); template for corrections

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7 Connections to Your Existing Resources

Geyer on Godambe: The Godambe information $G = HJ^{-1}H$ is exactly the object you need for profile composite likelihood inference. The sandwich variance G^{-1} replaces H^{-1} (Fisher information) everywhere. Geyer's treatment gives you the estimating-equations foundation from which composite likelihood asymptotics derive as a special case.

Jørgensen & Knudsen (Jantsura & Comet) on asymptotics: The asymptotic theory for composite likelihood estimators runs through the same central limit theorem for the composite score (a sum of dependent but asymptotically manageable terms), with the key twist being the Godambe rather than Fisher information.

Profile likelihood resources: The modified profile likelihood literature (Barndorff-Nielsen, Cox & Reid) is the closest structural analogue. The problems of $O(1)$ score bias and information bias from profiling are parallel; composite likelihood adds the additional complication of $H \neq J$, requiring the rescaling of ratio statistics described above.

The practical upshot: **profile the composite likelihood as you would any profile likelihood, then use Wald inference from the sandwich variance**, or apply the Geys–Molenberghs–Ryan rescaling to the ratio statistic. Do not use naive Wilks' theorem on the unscaled ratio.